

DETAILED EXPLANATION OF COMPUTER ARCHITECTURE FILE

CHAPTER 1 — INTRODUCTION TO COMPUTER ARCHITECTURE

Computer architecture describes how a computer is designed, how components connect, and how they work together. It has four areas: structure, organization, implementation, and performance. Architecture defines what the system can do (instruction set, data types). Organization defines how hardware executes architecture (control signals, memory technology).

CHAPTER 2 — STRUCTURE AND FUNCTION

A computer must process data, store data, move data, and control operations. Main components: CPU, main memory, I/O, system bus. CPU contains ALU (arithmetic), Control Unit (coordination), and registers (temporary storage).

CHAPTER 3 — VON NEUMANN ARCHITECTURE

Principles: instructions and data stored in same memory, addressed by numeric locations, executed sequentially. Important registers include MAR (address), MBR (data buffer), I/O registers. Memory stores instructions and data in numeric locations.

CHAPTER 4 — FUNCTIONAL COMPONENTS

Input unit (keyboard, scanner), CPU (ALU, CU, registers), memory (RAM), output unit (monitor, printer). CPU fetches, decodes, executes instructions. ALU performs arithmetic and logic. CU sends control signals. Registers store temporary values.

CHAPTER 5 — SYSTEM BUS

Bus = set of wires carrying signals. Address bus (unidirectional), data bus (bidirectional), control bus (timing and control signals). Address bus size determines how much memory can be accessed.

CHAPTER 6 — INTRODUCTION TO MICROPROCESSORS

Microprocessor = CPU on a chip. Processes binary instructions. Word sizes: 4 bits (nibble), 8 bits (byte), 16/32 bits (word). Larger word size = faster operations.

CHAPTER 7 — PROGRAMMING LANGUAGES

Levels: machine language (binary), assembly language (mnemonics), high-level languages (C, Java). Assembly has one-to-one relationship with machine code.

CHAPTER 8 — DATA REPRESENTATION

Binary, hexadecimal, number conversions, integer ranges, signed numbers using 2's complement. Character encoding: ASCII, Unicode. Signed numbers use MSB as sign bit.

CHAPTER 9 — MICROPROCESSOR ARCHITECTURE

IA■32 registers: EAX, EBX, ECX, EDX for arithmetic. EIP stores next instruction. EFLAGS stores status flags (zero, carry, overflow). Segment registers for memory segmentation.

CHAPTER 10 — MEMORY TECHNOLOGIES

ROM (permanent), EPROM (UV erasable), EEPROM (electrically erasable), FLASH (fast, used in SSD). RAM: DRAM (main memory), SRAM (cache).

CHAPTER 11 — DISK STORAGE

Access time = seek time + rotational latency + transfer time. Magnetic disks store data in tracks and sectors.

CHAPTER 12 — PIPELINING AND SUPERSCALAR

Pipelining overlaps multiple instruction stages. Superscalar processors (like Pentium) have multiple pipelines for parallel instruction execution.

INSTRUCTION EXECUTION CYCLE

Steps:

1. Fetch – read instruction from memory.
2. Decode – understand instruction type.
3. Operand Fetch – load needed operands.
4. Execute – ALU performs operation.
5. Write■back – store result to register or memory.

RELATABLE EXAMPLE

Like reading a to■do list:

Fetch = read the next task.

Decode = understand the task.

Operand fetch = gather materials.

Execute = perform the task.

Write■back = mark it completed.